

Work = Force * Distance**Hydrostatic Force = Density * Depth * Area****Mass = Density * Area****Moment = Distance * Mass****Center of Mass** $= (\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$

1. A spring has a natural length of 5 ft. A force of 4 lb stretches this spring 1/4 ft.
 (a) Find the work done in stretching this spring from its natural length to 7 ft.

$$\int_0^2 16x \, dx$$

- (b) Find the work done in stretching this spring from 8 ft to 9 ft.

$$\int_3^4 16x \, dx$$

2. How much work is done in emptying a cylindrical tank of radius 5 m and height 10 m by pumping the the water through a pipe extending 4 m above the tank. (Hint: Water weighs 9800 N/m³, or 62.4 lb/ft³.)

$$\int_0^{10} 9800\pi(5)^2(y+4) \, dy = \int_0^{10} 9800\pi(5)^2(14-y) \, dy = \int_4^{14} 9800\pi(5)^2 y \, dy$$

3. A rectangular gate built into a dam is 4 ft high and 8 ft wide. The top of the gate is 2 ft below the surface of the water. What is the force on the gate?

$$\int_0^4 62.4(8)(y+2) \, dy = \int_0^4 62.4(8)(6-y) \, dy = \int_2^6 62.4(8)(y) \, dy$$

4. What would be the force if the gate were the upper half of disc of radius 4 (and the top of the gate is 2 ft below the surface of the water)?

$$\int_0^4 62.4(6-y)\sqrt{4^2-y^2} \, dy$$

5. Find the mass, moments, and center of mass of a the triangular lamina bounded by the x - and y - axes and $y = 4 - 2x$. (As always, assume the density of this lamina to be 1.)

$$m = \int_0^2 4 - 2x \, dx = \int_0^4 \frac{1}{2} \left(\frac{4-y}{2} \right) \, dy$$

$$M_y = \int_0^2 x(1)(4-2x) \, dx = \int_0^4 \frac{1}{2} \left(\frac{4-y}{2} \right) (1) \left(\frac{4-y}{2} \right) \, dy$$

$$M_x = \int_0^2 \frac{1}{2}(4-2x)(1)(4-2x) \, dx = \int_0^4 y(1) \left(\frac{4-y}{2} \right) \, dy$$